PECAM-FY4A&B: Precipitation Estimation Using Chromatographic Analysis Method by Merging Enhanced Multispectral Infrared Observations From FengYun-4A and 4B

Siyu Zhu, Ziqiang Ma, Songkun Yan, Jintao Xu, and Xiaoqing Li

***Abstract*— Multiple spectral infrared (IR) observations onboard geostationary satellites are effective and widely used in estimating precipitation with high spatiotemporal resolutions. Currently, FengYun-4A (FY-4A) and FengYun-4B (FY-4B), representing the most advanced Chinese geostationary meteorological satellites, are equipped with Advanced Geosynchronous Radiation Imager (AGRI) to continuously observe the climate and weather over vast eastern Asia region. However, geographic factors, such as viewing zenith angle (VZA) and solar zenith angle (SZA), could result in systematic errors in estimating and merging precipitation. Therefore, motivated by analyzing the influence patterns of these physical factors on precipitation estimation, the precipitation estimation using the chromatographic analysis method by merging enhanced multispectral IR observations (PECAM) is proposed to generate the merged precipitation data covering observed fields of both FY-4A and FY-4B. The main conclusions are summarized as follows: 1) latitude, view zenith angle (VZA), and elevation exert a negative influence (up to 20 K) on averaged TBBs of single IR band (as $T_{10.8}$), causing precipitation overestimation; 2) $\Delta T_{7.1A-13.5A}$ and $\Delta T_{6.95B-13.3B}$, $\Delta T_{3.75H-13.5A}$ and $\Delta T_{3.75H-13.3B}$, and $\Delta T_{3.75H-3.75L}$ are mainly influenced by ecliptic obliquity angle (EOA), SZA, and shadow effect, causing seasonal patterns, diurnal fluctuations, and shadow effects, respectively; 3) compared to PERSIANN-CCS, FY4A-Official, and FY4B-Official, at hourly scale, PECAM-FY4A&B consistently outperforms them across CC, RMSE, and CSI metrics, with minimum improvements of approximately 0.046 in CC, 0.30 mm/h in RMSE, and 0.033 in CSI; and 4) meanwhile at daily scales, merged data from FY-4A&B shows overall improvements in CC, RMSE, and CSI, with at least 0.041, 0.80 mm/day, and 0.022, respectively. Foreseeably, the signal processing and merging strategy in PECAM has significant potential to serve as references for the estimation and integration of precipitation data from the FY-4 series, as well as the GOES, Meteosat, and Himawari series.**

***Index Terms*— Chromatographical analysis, data merging, FengYun-4A (FY-4A), FengYun-4B (FY-4B), precipitation estimation.**

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I. Introduction

PRECIPITATION (P), as one of the most important components of the water cycle, plays a crucial role in the changes in Earth's systems [1], [2]. Nowadays, multiple spectral infrared (IR) observations onboard geostationary satellites are effective and widely used in estimating precipitation with high spatiotemporal resolutions [3]. However, IR signals onboard geostationary (GEO) satellites are highly affected by geographical factors, such as solar zenith angle (SZA) and view zenith angle (VZA), which cause systematic errors in estimating precipitation [4], [5]. Meanwhile, in the present day, the potential benefits of multispectral IR data have not yet been fully explored, particularly when incorporating a deeper understanding of radiative transfer [4], [6]. Therefore, analyzing the influence patterns of these factors may provide better information for estimating precipitation using the Chinese meteorological satellites FengYun-4A and 4B [7], [8].

Precipitation estimation using IR observations from GEO satellites has evolved over several decades, beginning in the 1970s [9], [10], [11], [12], [13], [14]. The evolution of IR precipitation estimation and the previous research can be broadly categorized into three stages [4]. In the initial phase during the 1970s, precipitation estimates were predominantly constrained by the spatiotemporal resolutions of IR data obtained from GEO satellites. Meanwhile, statistical methods were extensively employed for precipitation estimation during this time [13], [14], [15]. After 1994, when the spatiotemporal resolution of IR observations from GEO satellites had a significant improvement, the resolutions of precipitation estimation also reached a higher level of about ∼1 hr and ∼4 km [16], [17]. With improvements in computer technology, machine learning (ML) has been introduced into precipitation estimating for automatically quantifying the relationships between

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precipitation and massive observed IR data (such as artificial neural network, ANN) [18], [19], [20]. Currently, benefiting from the new generations of GEO satellites with improvements in spectral resolutions, more advanced and more complex ML methods were applied to explore the potential of multispectral IR data [such as U-net and convolutional neural networks (CNNs)] [21], [22], [23], [24], [25]. Nevertheless, due to their low interpretability, ML-related methods, which often function as black-box models, constrain our ability to comprehend precipitation and complex information retrieval signals [26], [27]. To solve this problem, Zhu and Ma [4] proposed the precipitation estimation using chromatographical analysis (PECA) to quantify the relationships between precipitation and IR band differences, which supplied deeper perspectives on how the precipitation interacts with the different band features based on statistical and numerical relationships.

Standing as the representatives of cutting-edge Chinese meteorological GEO satellites, FengYun-4 series (including FengYun-4A and FengYun-4B currently) are equipped with the Advanced Geosynchronous Radiation Imager (AGRI), providing multispectral IR observations at high spatiotemporal resolutions [28], [29]. The official precipitation product is generated by matching the probability density function (pdf) matching in conjunction with IMERG and then establishes a corresponding relationship between the brightness temperature of one certain channel with the precipitation values, which has been continuously updated and improved in subsequent years [30]. ML methods, such as CNNs and random forest (RF), have become the prevailing approaches for precipitation estimation within the context of the FY-2 or FY-4 series [6], [31], [32], [33]. To gain a comprehensive understanding of the precipitation estimation process using multispectral IR bands, PECA-FY4A employed chromatographical theory to delineate the connections between precipitation values and specific band features of FY-4A/AGRI, resulting in notable enhancements in the overall evaluation metrics [4]. In the field of precipitation estimation, chromatographic theory was introduced under the assumption that signals and band features from satellite-based multispectral observations contain more information about precipitation. This allows us to quantify and utilize the relationships between band features and precipitation to generate more accurate estimations [4]. Based on the findings from PECA-FY4A, the multispectral IR bands of FY-4A/AGRI exhibit significant advantages in precipitation estimation. They are capable of distinguishing between various precipitation systems associated with typhoons, which is almost impossible for single bands. In addition, specific band features, such as the difference between bands 7 and 8 of FY-4A/AGRI, demonstrate unexpectedly strong responses to extreme precipitation events [4]. Furthermore, the band features of FY-4A/AGRI are adept at distinguishing various precipitation systems within convective rainfall, particularly those associated with frontal precipitation systems and the precipitation systems in which the IR signal from a single band is not sensitive to precipitation over land [34], [35]. Hence, it is imperative to take full advantage of AGRI onboard the FY-4 series to achieve improved precipitation estimation at high spatiotemporal resolutions while also enhancing our understanding of the intricate responses of complex precipitation to multispectral IR bands.

However, the aforementioned methods focus more on the statistical relationships between the precipitation and the IR signals, and the influence patterns of radiation conditions on IR bands and band differences have not been comprehensively investigated [36]. From the current studies, atmospheric corrections based on SZA are essential when retrieving parameters on the Earth's surface using near-IR bands, such as normalized difference vegetation index (NDVI) [37]. Meanwhile, the adjustment of SZA is also important when estimating precipitation based on IR observations. For example, the GOES Multispectral Rainfall Algorithm (GMSRA) primarily considers the SZA by adjusting the solar component in the 3.9-μm band using the cosine of the SZA, thereby enhancing the accuracy of rainfall detection from satellite data [38]. Zhuge et al. [39] consider the SZA by using a quasi-Lambertian surface adjustment (QLSA) method to normalize the visible albedo, taking into account the relative positions of the Sun, satellite, and clouds, thereby improving the accuracy of satellite precipitation retrievals. Prior to the multispectral era, bands around 10.8 μm were predominantly utilized as the primary features for estimating precipitation [19], [40]. ML methods are often seen as effective models capable of automatically accommodating all features [25], [32], [41], [42]. From the results of PECA, IR bands with shorter wavelengths would be more sensitive to solar radiation, and calculating band differences could partly solve this problem, such as band 7 minus band 8 (both around 3.75 μm) of FY-4A/AGRI [4]. Nevertheless, when considering other band features, such as the difference between band 7 ($\sim$3.75 μm) and band 14 ($\sim$13.5 μm) of FY-4A/AGRI, the SZA still leads to noticeable diurnal variations. This is primarily because band 7 experiences a significant increase in response to sunlight, while band 14 remains relatively stable under varying solar radiation conditions [34]. The diurnal fluctuations in band features will bring systematic errors in precipitation estimation, underscoring the importance and necessity of mitigating the impact of SZA [43]. In addition to SZA, factors, such as the VZA and ecliptic obliquity angle (EOA), can also influence IR band features from GEO satellites [44], [45]. From our perspective, applying the experienced equations to preprocess the band features is an effective and efficient way to mitigate the systematic errors caused by the various radiation conditions.

Another challenge when merging GEO satellite data is the issue of data discontinuity at the edges, where the data quality tends to degrade. This often results in geographic gaps in the merged data [4], [6], [46]. For instance, the state-of-the-science IR precipitation data PERSIANN-CCS combines data from the American GOES, the European Meteosat, and the Japanese Himawari satellites to produce global precipitation data [47], [48]. However, it still shows discontinuities in merged boundaries, meaning that more robust merging strategy is necessary for generating spatiotemporally continuous precipitation data with higher numerical consistence [4], [6]. Nevertheless, various complex merging strategies have been developed to improve the accuracies of precipitation data [46], [49], [50], [51], [52], [53], [54]. Zhu et al. [46] developed MASTMA to merge the multisourced precipitation data based on morphological features and optimal theory, using accessible features and Bayesian posterior probability

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to determine the reliabilities of precipitation records. In this process, more reliable/unreliable precipitation records would get higher/lower weights in the merging process. This principle is also applicable to other merging methods. For example, Lyu et al. [50] employed the triple collocation (TC) method to merge the precipitation data in regions lacking ground observations, where the weight of each input dataset is calculated as the product of the error variances of the other two datasets divided by the sum of the products of all pairwise error variances, ensuring optimized merging and minimized total error variance. After reviewing the core concepts of the aforementioned references, we recommend using a weighted averaging approach that combines both static weights (based on geographical location) and dynamic weights (based on precipitation patterns) to address the discontinuity issue.

Motivated by exploring the full potentials of multispectral IR bands in estimating precipitation and improving the precipitation accuracies based on Chinese FY-4 series, the Precipitation Estimation using Chromatographic Analysis method by merging enhanced multispectral IR observations (PECAM) of FengYun-4A&B based on radiative situations (PECAM-FY4A&B) is proposed. The most important objective of this study is to formulate innovative precipitation merging strategies for integrating precipitation data from FY-4A and FY-4B based on analyzing and discussing the impact of radiative conditions on the characteristics of AGRI multispectral IR bands in precipitation estimation [38], [39], [43], [55]. All endeavors in this study are aimed at unlocking the complete potential of geostationary satellites in the field of precipitation estimation.

II. Materials

A. Study Area

The study areas primarily encompass the observational domains of FY-4A and FY-4B satellites. The east–west boundaries extend approximately from 30°E to 152°W, while the north–south boundaries extend approximately from 75°N to 75°S [28], [56]. This expansive region encompasses a diverse range of surface environments and climates, including landmasses in China, India, Australia, parts of Central Asia, Southeast Asia, vast stretches of the Indian Ocean, Western Pacific, as well as various island regions such as Japan [4]. However, due to both the increased viewing angle and the curvature of the Earth, the sample density near the boundary is significantly lower than in the central region around the subpoint of a geostationary satellite, with the curvature of the Earth notably exacerbating this reduction in sample density. In this situation, moire fringes become evident in the boundary regions when the data are projected onto an equal latitude and longitude grid, which causes transformations of precipitation patterns [57]. Since the spatial resolution near the subpoint of AGRI onboard the GEO satellites can be as fine as approximately 4 km, while it degrades to around 10 km near the boundaries, we focus on regions within 65° from the subpoint for both FY-4A and FY-4B.

B. Data in This Study

1) Driven Data (FY-4A and FY-4B): The FY-4 series, developed and governed by the National Satellite Meteorological Centre (NSMC), stand as China's cutting-edge GEO meteorological satellites, providing substantial data for climate change analysis and disaster warning purposes [28]. FY-4A was launched on December 11, 2016 and finally located at 104.7°E, and it is primarily focused on covering China as its central area of interest. FY-4B was launched on June 3, 2021, and was initially positioned itself at approximately 123.5°E by June 10, 2021 [28], [58]. On April 11, 2021, FY-4B relocated to 133°E and has been operating from that position during the study period. On March 3, 2024, FY-4B drifted from 133°E to 105°E, taking over the functions of FY-4A, while FY-4A drifted from 104.7°E to 86.5°E. As well as GOES-R, Meteosat Second Generation (MSG), and Himawari-8, FY-4 series could be regarded as the third generation of GEO satellites due to its multispectral IR observation instruments named AGRI. From Table I, AGRI onboard FY-4A has a total of 14 bands, among which there are a total of eight IR bands varying from 3.75 to 13.5 μm. As for the AGRI onboard FY-4B which contains 15 bands in total, the improvements contain one extra band centered at 7.42 μm, and the water vapor band 7.1 μm has been replaced by 6.95 μm by NSMC. For FY-4A and FY-4B, the temperature of blackbody (TBB) data are accessible on the NSMC official website (http://satellite.nsmc.org.cn).

2) Training Data (GPM-DPR Precipitation Observation): The Global Precipitation Measurement (GPM) core observatory satellite was launched in February 2014 by the National Aeronautics and Space Administration (NASA), equipped with a dual-frequency precipitation radar (DPR) to enhance precipitation monitoring [2], [59]. DPR has the capability to detect vertical precipitation structures, thereby enabling more accurate precipitation estimation [60]. The surface precipitation rates from the normal scan of DPR version 6 are selected as the true values for training the model to take full advantage of its high accuracy and global scanning range. The DPR data are publically available at the NASA official website (https://disc.gsfc.nasa.gov).

3) Benchmark Data (PERSIANN-CCS and FY4QPE): The PERSIANN series represents state-of-the-science precipitation data primarily derived from IR observations from geostationary satellites [61]. Among these, PERSIANN-CCS stands out as one of the most well-known real-time global IR precipitation datasets [20]. It has been instrumental in global precipitation monitoring for nearly two decades and has played a vital role as a key component of IMERG [62]. Within the latitude range of 60°S to 60°N, PERSIANN-CCS aggregates the observations from GOES, Meteosat, and Himawari to generate global precipitation data at the resolutions of ~4 km and hourly. For downloading PERSIANN-CCS, the Center for Hydrometeorology and Remote Sensing (CHRS) Data Portal (http://chrsdata.eng.uci.edu/) is available.

The official precipitation products of FengYun-4 series (FY4-Official) are developed and released by NSMC, including the official precipitation data of FY-4A (FY4A-Official) and FY-4B (FY4B-Official). The resolutions depend on the scan modes of AGRI onboard the FY-4 series. Overall, the spatial resolutions are about ~4 km, which are the same as the input TBB of FY-4A&B, and the temporal resolutions are approximately 5 min/1 h for Northern Hemisphere (NH) mode/full disk (FD) mode, respectively [4], [6]. The

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TABLE I
BAND INFORMATION OF AGRI ONBOARD FY-4A AND FY-4B

FY4A		FY4B		Common		
Band No.	Wavelength (~ μm)	Band No.	Wavelength (~ μm)	Resolution (FY-4A, FY-4B) (~ km)	Main Applications	Symbols ($T_{\sim}$ or band ~)
1	0.47	**1**	0.47	1, 1	Aerosol	-
2	0.65	**2**	0.65	0.5 to 1, 0.5	Fog, cloud	-
3	0.825	**3**	0.825	1, 1	Vegetation	-
4	1.375	**4**	1.379	2, 2	Cirrus	-
5	1.61	**5**	1.61	2, 2	Cloud, snow	-
6	2.25	**6**	2.25	2 to 4, 2	Cirrus, aerosol	-
7	3.75 (high)	**7**	3.75 (high)	2. 2	Fire	$T_{3.75H}$
8	3.75 (low)	**8**	3.75 (low)	4, 4	Land surface	$T_{3.75L}$
9	6.25	**9**	6.25	4, 4	High-level WV	-
10	7.1	**10**	6.95	4, 4	Mid-level WV	$T_{7.1A}$&$T_{6.95B}$
-	-	**11**	7.42	-, 4	Low-level WV	-
11	8.5	**12**	8.55	4, 4	Cloud	-
12	10.8	**13**	10.8	4, 4	Surface temperature	$T_{10.8}$
13	12.0	**14**	12.0	4, 4	Surface temperature	-
14	13.5	**15**	13.3	4, 4	Cloud thickness	$T_{13.5A}$&$T_{13.3B}$

Note: Wavelength here represent the central wavelength of bands. WV represents water vapor. All the information is sourced from the official NSMC website, available in both Chinese and English versions. The primary applications refer to the intended uses of AGRI, although they serve additional purposes in precipitation estimation. Note that 3.75H and 3.75L represent the channel at 3.75μm taking the objectives with high albedo (e.g., cloud) and low albedo (e.g., land surface), at the spatial resolutions of 2 km and 4 km, with Noise Equivalent Delta Temperature (NEΔT) values as ≤0.7 K at 315 K and 0.2 K at 300 K, respectively.

approach for creating the official precipitation product for FY-4 involves aligning pdf data with IMERG to establish correlations between TBB of one certain channel and precipitation values [30]. This method has been continually updated and enhanced in recent years, leading to variations in the statistical characteristics of FY4-Official during different time periods. Similar to IR data from FY-4, the precipitation could also be found on NSMC official website (http://satellite.nsmc.org.cn).

4) Reference Data (IMERG): Developed by NASA, IMERG is recognized as one of the most advanced global precipitation datasets, integrating remote sensing observations from multiple sources to effectively harness the diverse strengths of various satellites [63]. In contrast to precipitation data derived solely from IR observations, IMERG places greater emphasis on microwave observations, resulting in improved accuracy of precipitation values and the depiction of precipitation system patterns [64], [65], [66]. Therefore, IMERG Final run Version 6 is regarded as the reference data to evaluate and analyze the precipitation patterns of IR-based precipitation estimations, especially in the gauge-sparse regions. In addition, DPR serves as a calibration reference for passive microwave (PMW) rain rate retrievals but is not directly used in IMERG. The IMERG series data can be downloaded from the NASA official website (https://disc.gsfc.nasa.gov).

5) Reference Data (Gauge): The gauge is widely recognized as the most accurate method for measuring precipitation values, due to its physical basis and direct measurement approach. In this study, precipitation measurements from Chinese meteorological gauges are used to evaluate the results of PECAM-FY4A&B at the hourly scale, which includes a total of around 69 000 gauges across Mainland China. Meanwhile, the data were downloaded from the NSMC official website (http://satellite.nsmc.org.cn). Meanwhile, a public dataset named Global Surface summary Of Day data (GSOD) is selected as ground reference data to evaluate the results of PECAM-FY4A&B [67]. It contains over 12 000 global meteorological gauge records and the data include precipitation, temperature, and so on. GSOD is developed by the National Oceanic and Atmospheric Administration (NOAA) and can be found and downloaded from the NOAA official website (https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily).

In this study, IMERG primarily serves to compare spatial precipitation patterns of specific events over oceanic regions. Meanwhile, IMERG has documented shortcomings, such as underestimating precipitation in high mountain areas during cold seasons [68], [69]. Consequently, gauges serve as the definitive reference data for quantifying the performance of PECAM-FY4A&B and for metric calculation. Furthermore, since DPR data were employed in training our model, it is unsuitable as reference data to ensure the independence of our evaluation.

III. METHODOLOGY

A. Algorithm Structure of PECAM-FY4A&B

PECA was originally designed for excavating the potential effective information from the multispectral IR signals of FY-4A/AGRI, and this strategy also works for FY-4B/AGRI. However, the band features selected in PECA have significant systematic biases due to different atmospheric and geometric effects. Compared to PECA, PECAM stands more for both multispectral and merging. As the upgraded version of PECA, it fully considers the influence of radiative conditions on precipitation estimation and merging. By introducing radiative theory, PECAM not only enhances the signal-to-noise ratio (SNR) and reduces the systematic errors of band features but also better alleviates data discontinuities in precipitation merging through weighted adjustments. As a result, PECAM is capable of generating more accurate precipitation data that cover the observational ranges of FY-4A and FY-4B, and it also provides better interpretation and understanding of the radiative transfer processes in multispectral IR signals.

The algorithm structure has six main steps (Fig. 1): 1) data preparation; 2) feature enhancements; 3) determination of P events; 4) estimation of P values; 5) fusion of *P* events and

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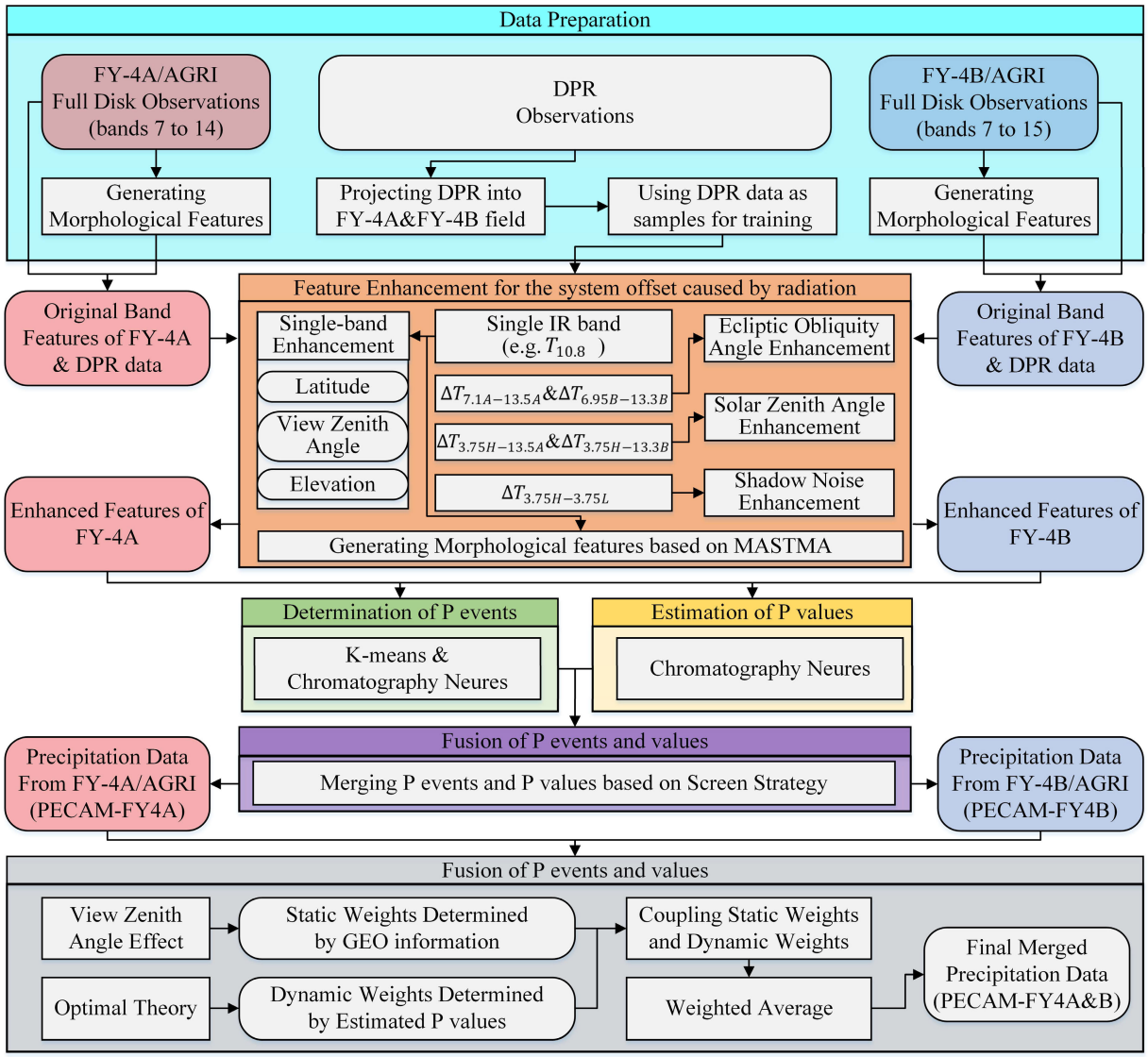


Fig. 1. Flowchart of the PECAM-FY4A&B.

P values; and 6) merging of results from FY-4A and FY-4B. Among them, step 5 adopts the same structure as PECA; meanwhile, steps 1, 3, and 4 are all upgraded to match the band features after feature enhancements [4]. Inspired by Zhu and Ma [35] and Zhu et al. [46], steps 2 and 5 are first proposed to solve the systematic-fluctuation problem caused by different radiative situations and discontinuity problem in the merging process, respectively. Meanwhile, we match the DPR and AGRI data by aligning pixel centers based on the nearest distance at both spatial and temporal dimensions, to ensure the integrity of the training samples.

In PECAM, precipitation estimation mainly comprises two independent procedures for determining the final precipitation: determining P events and estimating P values. When determining P events, all bands of FY-4A and FY-4B satellites are considered. This comprehensive approach leverages the full range of band information to improve the accuracy of precipitation detection. In this structure, chromatographical neurons are defined as functional modules that take band features/precipitation estimation as input/output, respectively, and then process them through the specifically designed activation functions (e.g., sigmoid), which align with the pattern curves between band features and precipitation [4].

When estimating precipitation events, the band features are selected based on the following principles: 1) they should have the highest possible correlation with precipitation and 2) they should have the lowest correlation with each other (e.g., linear regression) [4]. Therefore, only certain band features that have less redundant information with each other are selected to avoid overfitting in this model. In addition, the gauges are matched with the precipitation pixels that contain the gauges' locations for both hourly and daily evaluations.

B. Feature Enhancements for Multispectral Signals From PECAM-FY4A&B

The original PECA structure mainly focuses on exploring the relationships between direct band differences and precipitation and then analyzing the characteristic responses of these band features by considering their radiative features. However, the original band differences suffer from the systematic biases caused by different radiative situations, which in turn impacts the direct relationship between IR signals and precipitation.

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TABLE II
MAIN RADIATIVE INFLUENCE FACTORS CONSIDERED FOR SELECTED BAND FEATURES

Radiative Influence Factors	$T_{10.8}$	$\Delta T_{7.1A-13.5A}$ & $\Delta T_{6.95B-13.3B}$	$\Delta T_{3.75H-13.5A}$ & $\Delta T_{3.75H-13.3B}$	$\Delta T_{3.75H-3.75L}$
Latitude	Major	Medium	Medium	Slight
View Zenith Angle	Major	Medium	Medium	Slight
Elevation	Major	Medium	Medium	Slight
Solar Zenith Angle	Slight	Slight	Major	Medium
Ecliptic Obliquity Angle	Slight	Major	Medium	Slight
Shadow Effect	Slight	Slight	Slight	Major

Major, Medium, and Slight represent that certain radiative influence factors could cause significant, general, and unapparent systematic bias to certain band features.

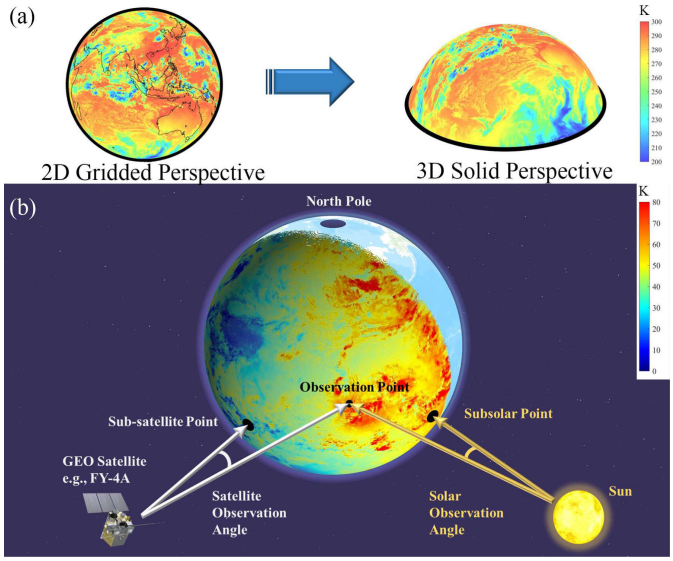


Fig. 2. Diagram of feature enhancements, including (a) improvements on perspectives from 2-D to 3-D and (b) taking band 3.75H μm minus band 13.5A μm as an example to show the systematic influence of solar radiations on band features.

Compared to PECA, PECAM enhances the approach by interpreting data through three dimensions rather than two dimensions, incorporating factors such as the VZA and SZA [see Fig. 2(a)]. Although this adjustment might not greatly improve the accuracy of precipitation measurements, it is a valuable and progressive step in IR precipitation estimation. We have concluded the main influence patterns of radiative situations for different band features in Table II, and the systematic biases caused by them have been adjusted, by quantifying the complex radiative transfer process into mathematic functions with calibrated parameters.

Single-band enhancement includes the systematic adjustments of single IR band based on latitude (Lat), VZA, and elevation (Ele). Latitude represents the climate properties of different regions, including averaged temperature and atmosphere circulation pattern. IR radiation emission of Earth is highly related to VZA, which would cause the decrease of observed TBB over the edge regions of the view field (e.g., polar areas). As a result, lower TBB at high latitude would indicate higher rain rates, which could cause severe overestimations over these regions. Meanwhile, higher VZA and elevation can also cause lower TBBs for IR bands in clear skies, making it necessary to avoid misrecognition and overestimation of precipitation in these situations [43]. Therefore, we make a cosine-based compensation to the influence of Lat, VZA, and Ele. The patterns of (1)–(3) correspond to the patterns of averaged changing curves (P values as y and Lat, VZA, and Ele as x) for nonprecipitation pixels. Afterward, the final enhanced feature is generated as shown in (4). Equations and parameters in (1)–(4) are designed to roughly mitigate the systematic errors caused by latitude, VZA, and elevation. These equations are designed to best fit the patterns of changing curves, using brightness temperature (y) of no-precipitation events and geographical features (x) as variables. The nonlinear least-squares method is used to optimize the shape of the fitting curves. Similarly, the equations and parameters for band features are determined by initially selecting an equation format based on experience, followed by applying the nonlinear least-squares method to no-precipitation samples with DPR as the reference data

$$T'_{\mathrm{Lat}} = T_{\mathrm{IR}} \cdot (1 - \cos(\mathrm{abs}(\mathrm{Lat}_0 - \mathrm{Lat}_{\mathrm{subsolar}}))) \cdot \mathrm{Para}_1 \quad (1)$$

$$T'_{\mathrm{VZA}} = T_{\mathrm{IR}} \cdot (1 - \cos(\mathrm{VZA})) \cdot \mathrm{Para}_2 \quad (2)$$

$$T'_{\mathrm{Ele}} = T_{\mathrm{IR}} \cdot (1 - \cos(\mathrm{Ele}/\mathrm{Para}_4)) \cdot \mathrm{Para}_3 \quad (3)$$

$$\mathrm{ProBF}_1 = T_{\mathrm{IR}} + T'_{\mathrm{Lat}} + T'_{\mathrm{VZA}} + T'_{\mathrm{Ele}} \quad (4)$$

where Lat_0 and $\mathrm{Lat}_{\mathrm{subsolar}}$ represent the latitude of current point and subsolar point, respectively; VZA and Ele denote the VZA and elevation, respectively; T_{IR} and ProBF_1 indicate the original observed TBBs and the TBBs after feature enhancements to drive the model, respectively, during which T'_{Lat}, T'_{VZA}, and T'_{Ele} represent the enhanced parts of TBBs caused by Lat, VZA, and Ele, respectively; and Para_1–Para_4 are the static parameters that are set for PECAM-FY4A&B as 0.08, 0.03, 0.03, and 4000, respectively.

Ecliptic obliquity angle (EOA, the tilt of the Earth's axis relative to its orbit around the Sun) has significant relationships with the climates and seasons, especially the atmospheric temperatures at different latitudes. Taking FY-4A as an example, $T_{13.5A}$ (corresponding to $T_{13.5A}$ of FY-4B) is more sensitive to atmospheric temperature, and thus, $\Delta T_{7.1A-13.5A}$ (for FY-4B, it is $\Delta T_{6.95B-13.3B}$) shows extremely high values over the cold and rainless regions, resulting in substantial no-rain contamination. Due to the high $\Delta T_{7.1A-13.5A}$ signals with no precipitation caused by EOA, EOA enhancement (5) is implemented to extract the extreme precipitation signal by combining normalized high $\Delta T_{7.1A-13.5A}$ and low $T_{10.8}$, in which case the fluctuations caused by EOA in $\Delta T_{7.1A-13.5A}$

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will be significantly mitigated

$$\mathrm{ProBF}_2 = (\Delta T_{7.1A-13.5A} - \mu_1)/\sigma_1 \cdot (T_{10.8} - \mu_2)/\sigma_2 \quad (5)$$

where $\Delta T_{7.1A-13.5A}$ and $T_{10.8}$ are the band differences between bands 10 and 14, and TBBs of band 12 of FY-4A/AGRI, respectively; ProBF_2 is the band feature after EOA enhancement, which is used to replace the original $\Delta T_{7.1A-13.5A}$; and μ_1, σ_1, μ_2, and σ_2 are all parameters for standardizing $\Delta T_{7.1A-13.5A}$ and T_{12}, which are assigned as 0, 5, 280, and 20, respectively.

The SZA enhancement is designed to correct systematic errors arising from varying solar conditions, as the same precipitation system can exhibit different IR signal characteristics under different solar radiation conditions [39]. IR bands with shorter wavelengths are more sensitive to solar radiation, and thus, bands 7 and 8 of FY-4A&B/AGRI have significant periodic changing patterns with the periodic solar radiations. Taking $\Delta T_{3.75H-13.5A}$ (for FY-4B, it is $\Delta T_{3.75H-13.3B}$) as an example, this quantity becomes higher when precipitation occurs by daylight but becomes lower over the precipitation regions at night. Therefore, changing patterns of averaged TBBs at clear sky over the FD were quantified as the baseline to calibrate the relationship between the predictors from AGRI and rainfall rate, and the bias from the baseline is regarded as the indicator of clouds and precipitation. For $\Delta T_{3.75H-13.5A}$, the precipitation region could be effectively extracted after applying SZA enhancement (6)–(7)

$$\mathrm{Ave}T_{3.75H-13.5A} = \mathrm{Para}_1(1 - \mathrm{ReLU}(\cos(\mathrm{SZA})))^4 + \mathrm{Para}_2 \quad (6)$$

$$\mathrm{ProBF}_3 = \Delta T_{3.75H-13.5A} - \mathrm{Ave}T_{3.75H-13.5A} \quad (7)$$

where $\Delta T_{3.75H-13.5A}$ represents the band differences between bands 7 and 14 of FY-4A/AGRI; $\mathrm{Ave}T_{3.75H-13.5A}$ is the averaged TBBs at clear sky for $\Delta T_{3.75H-13.5A}$; SZA denotes the SZA for every considered pixel at certain time; Para_1 and Para_2 are the parameters in generating $\mathrm{Ave}T_{3.75H-13.5A}$, which are set as -18 and 46, respectively; and ProBF_3 is the enhanced band feature of $\Delta T_{3.75H-13.5A}$.

The shadow effect only occurs in $\Delta T_{3.75H-3.75L}$ among all IR band features of PECA. Ba and Gruber [38] have extracted the emitted component for the 3.9-μm band by estimating its reflected component, illustrating that the TBBs of bands around this wavelength are influenced by both reflection and emission. Because band 3.75L is more sensitive to precipitation than 3.75H (see Table I), $\Delta T_{3.75H-3.75L}$ could significantly respond to extreme precipitation, especially at night. However, due to the intersection angle between SZA and VZA, the shadow of precipitation systems and clouds occurs in distribution maps of $\Delta T_{3.75H-3.75L}$, shown as the extremely low values around the precipitation systems. Therefore, the ReLU function (8) and a 5×5 filter (9) were applied to remove the shadow and enhance the signal of precipitation. We selected a 5×5 filter as it effectively captures the features of the bands and maximizes noise reduction

$$\mathrm{MidBF}_4 = \mathrm{ReLU}(\Delta T_{3.75H-3.75L}) \quad (8)$$

$$\mathrm{ProBF}_4 = \mathrm{filter}(\mathrm{MidBF}_{3.75H-3.75L}) \quad (9)$$

where ReLU is a function with threshold as zero for removing the negative values; MidBF_{7-8} and ProBF_4 are the middle values and final enhanced band features of $\Delta T_{3.75H-3.75L}$, respectively; and filter means to apply averaging sliding window filtering with a window of 5×5 size.

In addition, the morphological features are calculated based on the single 10.8-μm band due to its wide application in IR precipitation estimation. These features are introduced in precipitation estimation or merging processes to extract information about the spatial and temporal patterns of precipitation shapes [46]. They are integer values that indicate the distance from the centers of precipitation events, with only positive morphological features considered in this study. Positive integers represent proximity to the precipitation event center, with larger numbers indicating closer proximity to the center. The thresholds for the precipitation events are determined as 220, 235, and 253 K based on the methodology of PERSIANN-CCS [20]. In this section, the P value and P event refer to the PECA [4]. Meanwhile, nonprecipitating pixels are selected to calibrate the relationships in order to reduce the systematic errors, which will be influenced by surface radiation although in clear sky.

C. Merging Strategies From PECAM-FY4A&B

Merging the precipitation data from different GEO satellites is an effective approach to generate the data with broader coverage. As shown in Fig. 3, after FY-4B changed its location on April 11, 2022, FY-4A and FY-4B are currently located at 104.7°E and 133°E, respectively. Therefore, by considering FY-4B, the west boundary of PECA-FY4A could be extended approximately 28° from the original ~180°E to about −152°W. However, due to the different positions of FY-4A and FY-4B, when they observe the same location, the observations between the two satellites differ, resulting in inconsistent error levels in precipitation estimation. From Fig. 3(b), it is evident that the deviations of GEO satellites exhibit positive correlations with VZA and the height of the top clouds. This implies that the deviations tend to be more significant when farther from the subpoint of the GEO satellite. To determine VZA [see Fig. 3(c)], location geocentric angle (LGA) and satellite observation angle (SOA) could be added to get VZA. In this process, LGA is directly related to the longitude of surface point and SOA could be calculated by a trigonometric function. Because geostationary satellite height (GSH) is about 5.6 times Earth's radius, VZA closely approximates LGA and they are in a one-to-one correspondence (surjective mapping in math). Consequently, they exhibit similar patterns and have comparable impacts on band features.

From the aforementioned properties of GEO satellites, higher VZA leads to larger deviations, which means that the observations are more unreliable over the view field margin than the central regions. For example, in boundary regions, the emitted and reflected radiation from clouds is affected by the increased path length and scattering effects at higher viewing zenith angles (VZAs), leading to greater uncertainties in the observations. Vicente et al. [55] present a comprehensive study on parallax correction, showcasing significant improvements in the localization and quantification of precipitation, especially in regions distant from the satellite's subpoint. This research details the use of Cartesian coordinate transformations and precise calculations to correct apparent cloud-top positions,

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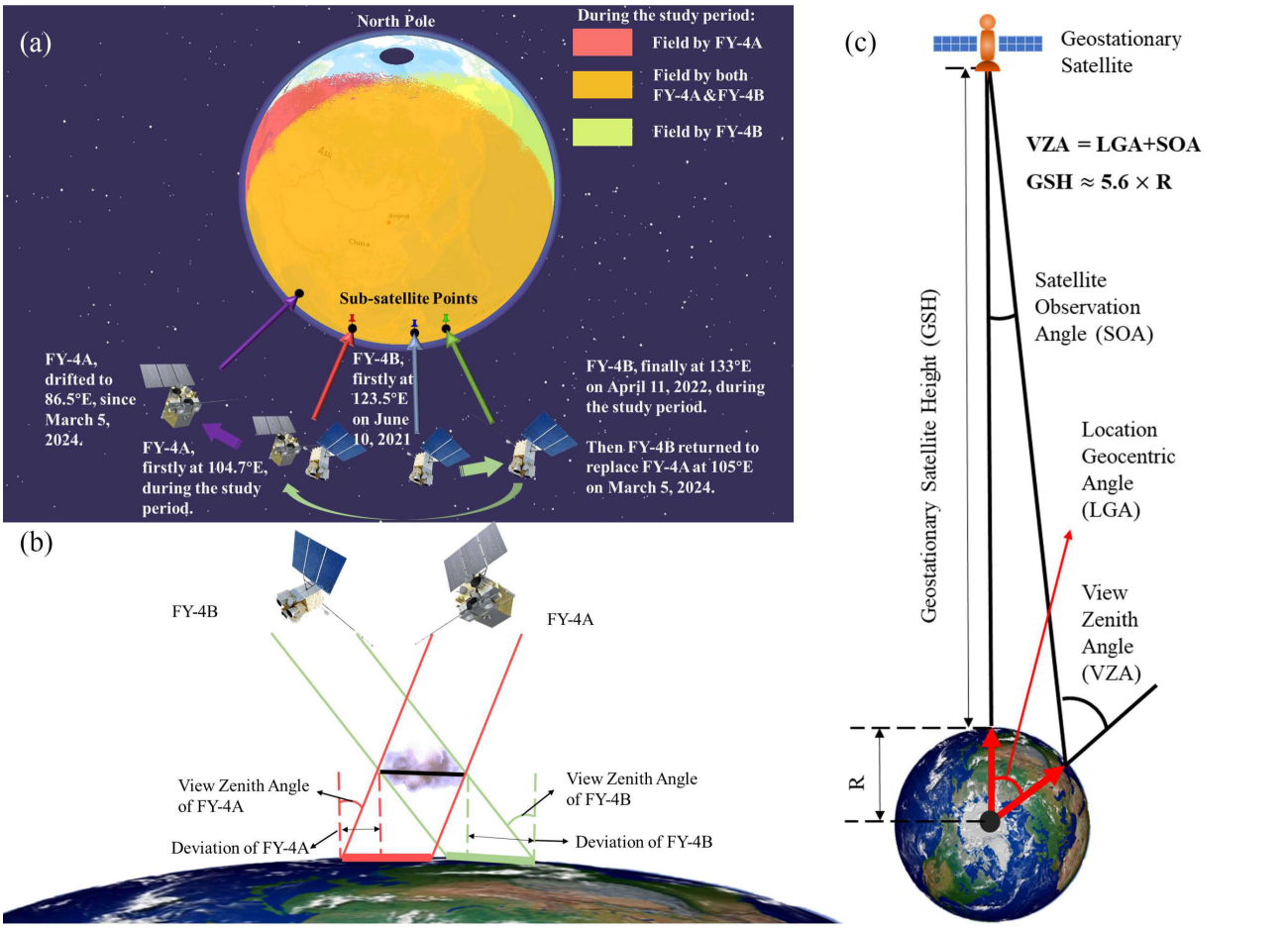


Fig. 3. Diagram of basic information for merging process, including (a) locations and corresponding coverages of FY-4A&B, (b) observing deviations of FY-4A&B due to the differences in view zenith angles, and (c) relative locations between single geostationary satellites and Earth. Please note that Fig. 3(b) is the schematic showing the deviation caused by the view zenith angle.

thereby enhancing the accuracy of real-time rainfall rate estimations. Motivated by their findings, we applied static and dynamic weights to optimize the parallax correction process, effectively mitigating systematic bias caused by varying VZA. Static weights are proposed to reflect the reliabilities of different locations as shown in (10), which range from 0 to 1 as a cosine function of the distance from the subpoint. As for the dynamic weights, it is assumed that higher precipitation values should be given more consideration under optimal theory. Therefore, (11) is applied to calculate dynamic weights with the control parameter K, which is empirically determined to scale precipitation in dynamic weight calculations to match static weight calculations. The primary role of K is to refine the spatial distribution of weights during the data merging process, which helps to adjust the numerical distribution of precipitation values to a more reasonable range for exponential analysis, and thus, it is assigned based on our experiences. Afterward, add static weights and dynamic weights to generate final weights, which are then utilized in the merging process for weight averaging. It is worth explaining that the more accurate approach would be to move the pixels closer to the satellite subpoint based on the height of the precipitation clouds. However, due to the uncertainties in estimating cloud heights and for the sake of efficiency, we have adopted a simpler method based on adjusted weights

$$\mathrm{SW} = \mathrm{Relu}(\cos(\mathrm{distance}(\mathrm{Pixel}, \mathrm{Subpoint}))) \tag{10}$$

$$\mathrm{DW} = \frac{1 - 1/\exp(\mathrm{PV}/K)}{1 + 1/\exp(\mathrm{PV}/K)} \tag{11}$$

$$\mathrm{FW} = \mathrm{DW} + \mathrm{SW} \tag{12}$$

where SW, DW, and FW represent static weights, dynamic weights, and final weights, respectively; distance is the function for calculating the surface distance between two different locations on the Earth; Pixel and Subpoint denote the location of current pixel and the subpoint of the satellite, respectively; PV is the precipitation value recorded in merged data; and K is a parameter in merging process, which is assigned as 5 mm/h to adjust the numerical distribution of precipitation values that is set based on our experience.

IV. Results

A. Effects of Feature Enhancements for Main Signal Features in PECA

In IR precipitation estimation, $T_{10.8}$ is predominately considered because it shows the significant response to precipitation and is relatively insensitive to solar radiation. Taking four scenarios on August 3, 2022 as an example (similarly below), $T_{10.8}$ could capture precipitation based on lower TBBs. However, because of the higher VZA around the observation field of FY-4A, the averaged TBBs at the boundary regions are lower than the central TBBs up to 20 K without considering precipitation. If using the same estimation strategies for all

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regions, the precipitation at the boundaries will be systematically overestimated. Therefore, in PECAM, we decided to increase the TBBs over the boundary regions as mentioned in Section III to reduce this systematic error. Dividing the estimation strategies for different regions is limited by insufficient samples and the requirement of continuity. As shown in Fig. 4, the TBBs of original band features [see Fig. 4(a)–(d)] for precipitation/clear-sky regions are ∼230 and ∼260 K, which are increased in enhanced band features [see Fig. 4(e)–(h)] into ∼250 and ∼280 K. This enhancement is predicated on the assumption that, due to the similar IR emissions from the Earth's surface and clouds, the VZA gradually increases from the center to the edge of the observation field, leading to a gradual attenuation of the radiation received by the IR sensor onboard GEO satellites. In addition, the degree of attenuation also depends on the cloud-top height, as the actual amount of atmosphere above the cloud top affects the observed brightness temperatures [43].

Different from $T_{10.8}$, $T_{7.1A-13.5A}$ has significant seasonal patterns, which are controlled by EOA (see Fig. 5). On August 3, the original $T_{7.1A-13.5A}$ shows significantly high values over the vast regions of the Southern Hemisphere (over 8 K), while over the regions of the North Hemisphere, TBBs with no precipitation (e.g., Tibetan Plateau and Central Pacific) are lower than 2 K. Not coincidentally, $\Delta T_{7.1A-13.5A}$ tends to be high (∼4 K) over precipitation regions in the NH, such as the Northern Indian Ocean and Northwest Pacific. In the Southern Hemisphere, however, extremely high $\Delta T_{7.1A-13.5A}$ values, driven by lower surface and atmospheric temperatures at higher latitudes causing lower $T_{13.5A}$, may be confused with precipitation-related signals near the equator. This misinterpretation can lead to overestimations in precipitation recognition, particularly in regions with lower surface and atmospheric temperatures (e.g., areas near the poles). The $T_{13.5A}$ channel is sensitive to cloud-top temperature, making it effective for reflecting and retrieving the atmospheric temperature over the cloud top. In this context, both the vertical atmospheric temperature profile and water vapor content significantly influence $T_{7.1A-13.5A}$. During extreme precipitation events, an increase in $T_{7.1A-13.5A}$ is observed as a characteristic feature. After extracting useful features, the enhanced band features can capture the extreme precipitation with high values (over 1), and the features over the regions that are not prone to precipitation have been removed.

As depicted in Fig. 6(a)–(d), solar radiation has the potential to cause a substantial rise in $T_{3.75H-13.5A}$, consequently giving rise to pronounced diurnal fluctuations, which are highly related to SZA. In this situation, precipitation regions (e.g., Northern Indian Ocean) show higher values of $T_{3.75H-13.5A}$ than surrounding areas, while the averaged values of $T_{3.75H-13.5A}$ during the day is much higher (up to 40 K) than the values at night. Because $T_{3.75H}$ is highly sensitive to solar radiation, while $T_{13.5A}$ is not, the patterns of $T_{3.75H-13.5A}$ are mainly caused by the diurnal fluctuations of $T_{3.75H}$, which is more obvious when subtracting a background baseline such as $T_{13.5A}$. Therefore, we used a parameter-controlled cosine-based function to simulate this systematic changing caused by solar radiation. After removing this offset, the enhanced band features [see Fig. 6(e)–(h)] demonstrate more direct relationships with precipitation, making them more readily distinguishable

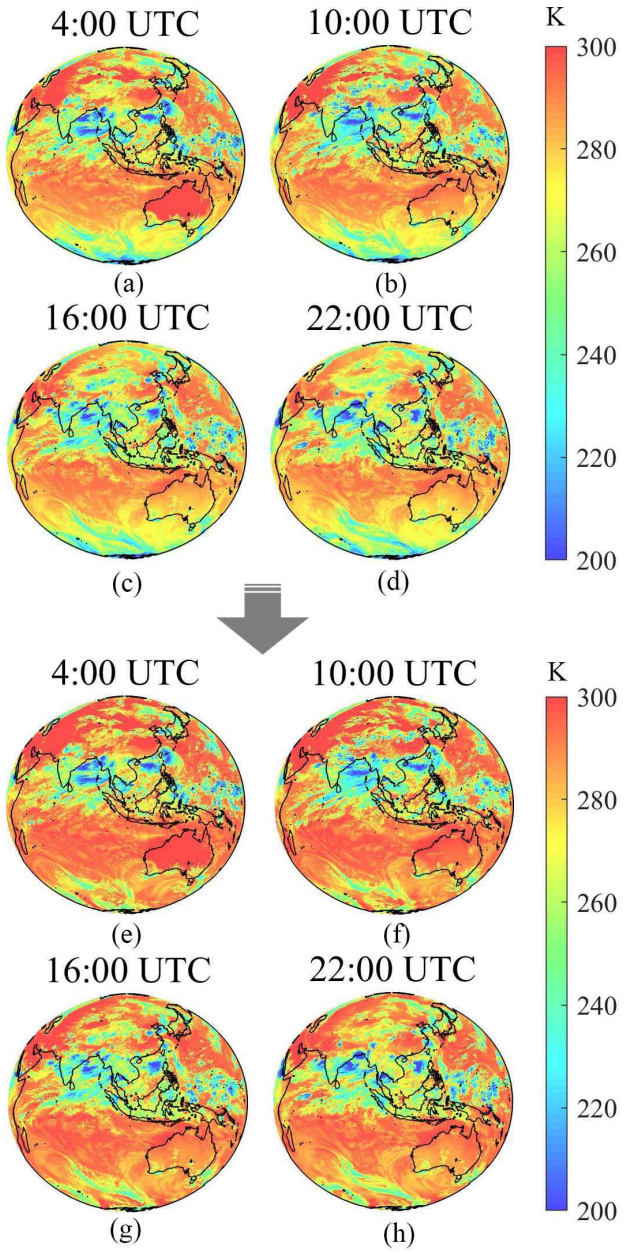


Fig. 4. Effect of feature enhancements for band 10.8 μm of FY-4A/AGRI on August 3, 2022, including (a)–(d) original band features and (e)–(h) enhanced band features.

from the complex radiation backgrounds. In addition, the signal of $T_{3.75H-13.5A}$ for precipitation is more pronounced in the central areas of satellite views compared to the boundary areas.

In PECA, $T_{3.75H-3.75L}$ has been proven to capture extreme precipitation due to its strong response, especially for the extreme precipitation over the eye wall regions of typhoons. However, the shadow effects (lower than −3 K) extensively exist around the precipitation regions (e.g., the events over India, higher than 5 K), which bring many random errors in quantifying the relationships between precipitation and $T_{3.75H-3.75L}$ [see Fig. 7(a)–(d)]. The shadow effects are related to the SZA, the patterns of which are difficult to quantify. Therefore, the noise is simply removed by the ReLU function, which has been proved to be effective and efficient. The

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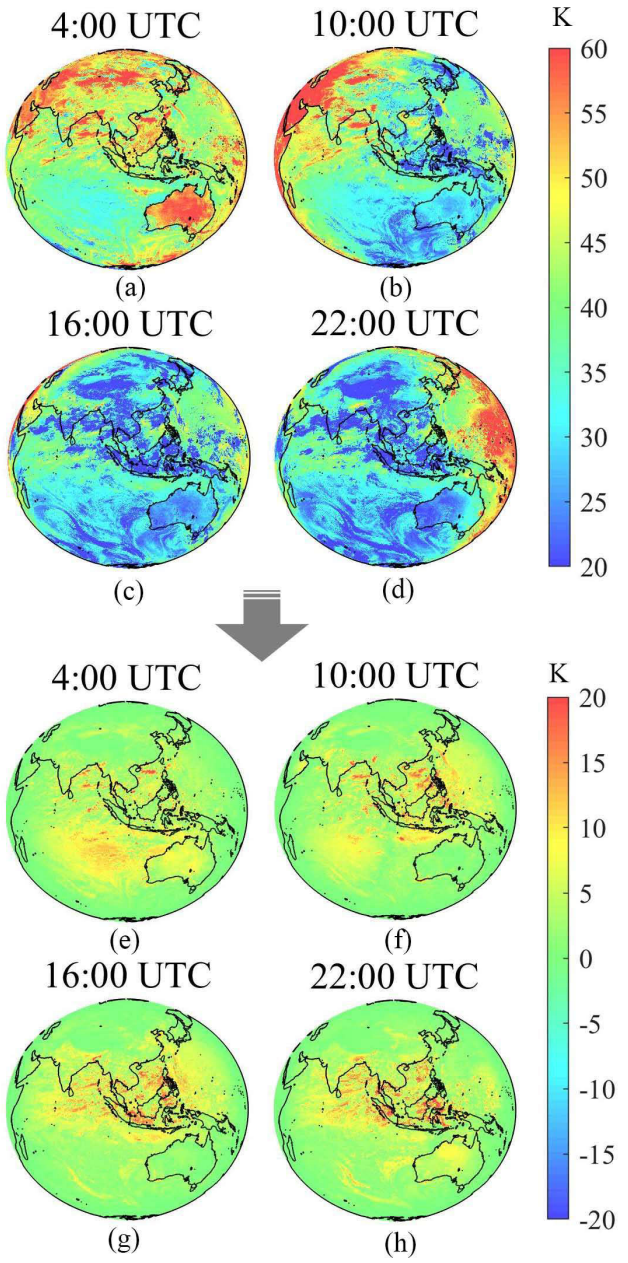


Fig. 5. Effect of feature enhancements for band 7.1A μm minus band 13.5A μm of FY-4A/AGRI on August 3, 2022, including (a)–(d) original band features and (e)–(h) enhanced band features.

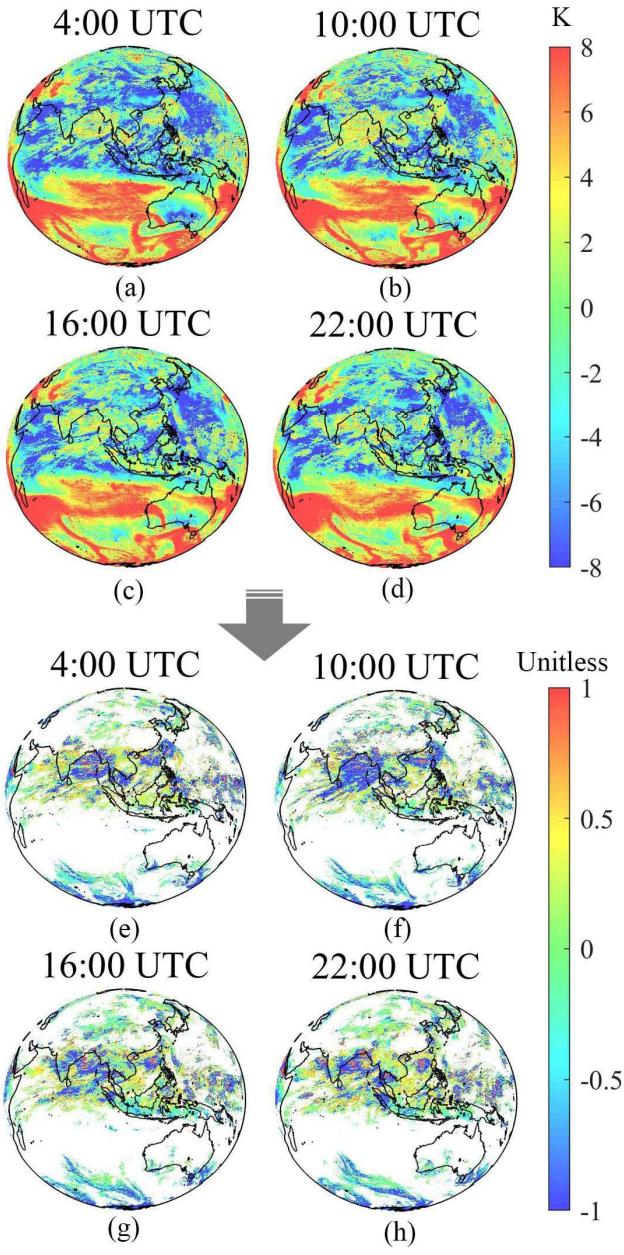


Fig. 6. Effect of feature enhancements for band 3.75H μm minus band 13.5A μm of FY-4A/AGRI on August 3, 2022, including (a)–(d) original band features and (e)–(h) enhanced band features.

enhanced band features [see Fig. 7(e)–(h)] are capable of better highlighting the precipitation, which is more obvious at night without solar radiation. By enhancing the band features, the relationships between band features and precipitation could be better revealed, contributing to our understanding of precipitation patterns in IR signals.

B. General Spatial Patterns of PECAM-FY4A&B and Other Precipitation Products

To analyze the performances of PECAM-FY4A&B, the cumulative precipitation has been shown in Fig. 8 with other precipitation products, including PERSIANN-CCS, IMERG-LR, FY4A-Official, and FY4B-Official. When projected into equal-latitude–longitude grids, we assign higher weights to regions near the subpoints of FY-4A and FY-4B compared to the boundary regions in the merging process [see Fig. 8(a)]. The sample points of AGRI become sparser from central regions to edges, which results in a Moire fringe loop (MFL), a discontinuous projection region around the edges. The data in the MFL cannot show complete precipitation patterns here and thus is excluded. Afterward, PECAM-FY4A&B [see Fig. 8(b)] shows higher consistencies with IMERG than PERSIANN-CCS [see Fig. 8(d)], and it could reflect more regions with intense precipitation [e.g., the northern Indian Ocean, the East China Sea, and Southeast Asia, Fig. 8(c)]. Meanwhile, FY4A-Official shows overestimation at the boundary areas (such as ~160°E and ~45°N, and ~160°E and ~30°S). Similarly, FY4B-Official

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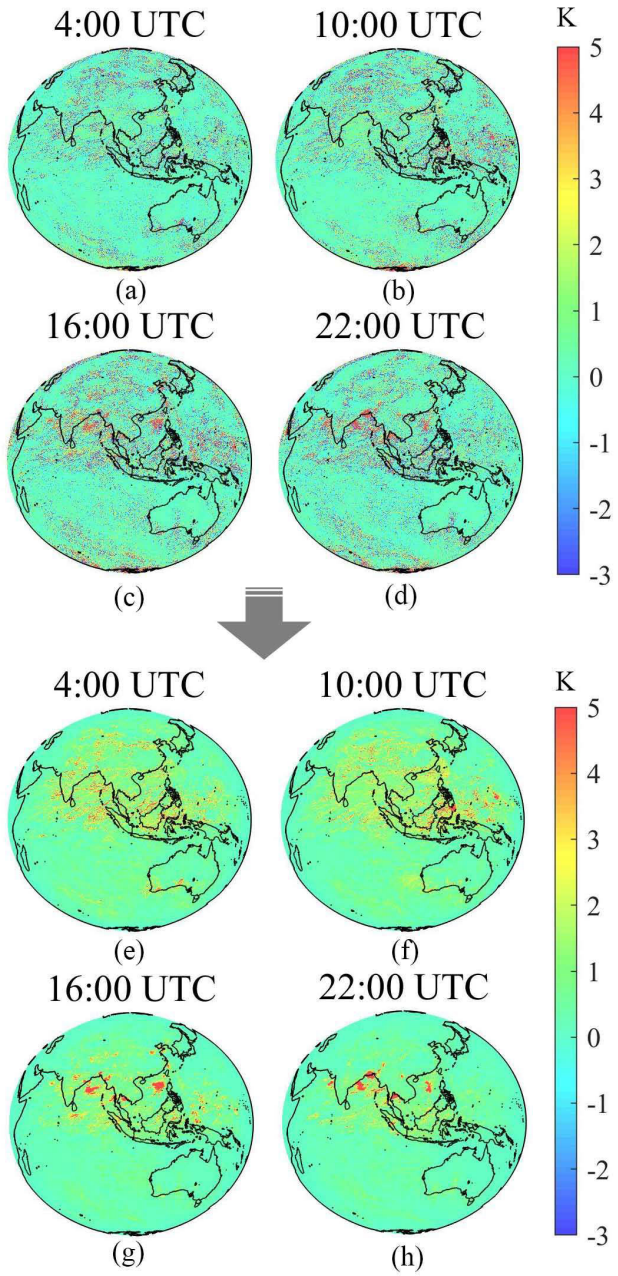


Fig. 7. Effect of feature enhancements for band 3.75H μm minus band 3.75L μm of FY-4A/AGRI on August 3, 2022, including (a)–(d) original band features and (e)–(h) enhanced band features.

exhibits severe overestimates over the region of ∼160°E and ∼30°S. The overestimations may arise due to the previously mentioned occurrence of low TBBs in the boundary regions. This underscores the necessity for enhancing the features of IR signals from FY-4A&B/AGRI. Notably, FY4B-Official more accurately represents precipitation patterns over the central regions compared to FY4A-Official, yet it still misses some areas of high-intensity precipitation, such as the region around 150°E and 30°N.

C. Evaluations of PECAM-FY4A&B, CCS, FY4A-Official, and FY4B-Official Based on Chinese Hourly Gauges

The correlation coefficient (CC), root-mean-square error (RMSE), and critical success index (CSI) are well-established metrics for evaluating precipitation products. CC and RMSE measure the continuity of precipitation values, whereas CSI assesses the ability to detect precipitation events (defined as precipitation exceeding 0.1 mm/h).

In hourly assessments, CC, RMSE, and CSI are calculated using data from around 69 000 Chinese meteorological gauges. Analysis of Fig. 9(a) reveals that the CC for PECAM-FY4A&B significantly surpasses that of other precipitation products, particularly in Southeast Mainland China, where it consistently exceeds 0.25 across a broad area. This observation is numerically supported by the box plot in Fig. 9(m), which shows that PECAM-FY4A&B (∼0.230) outperforms CCS (∼0.122), FY4A-Official (∼0.177), and FY4B-Official (∼0.184). Regarding RMSE, PECAM-FY4A&B and CCS both exhibit a trend of decreasing values from the south to the north and from coastal to inland areas. Conversely, FY4A-Official and FY4B-Official demonstrate greater uncertainties (RMSE over 2 mm/h) along the eastern coastline. The average RMSE for PECAM-FY4A&B is notably lower at ∼1.01 mm/h, compared to ∼1.32 mm/h for CCS, ∼1.49 mm/h for FY4A-Official, and ∼1.84 mm/h for FY4B-Official. As the indicator of precision in capturing precipitation events, CSI also underscores the superiority of PECAM-FY4A&B. It shows substantial enhancements in Central China, particularly in Sichuan and Shaanxi Provinces, as well as in the Northeast, including Liaoning and Jilin provinces. The average CSI for PECAM-FY4A&B (∼0.190) significantly exceeds those of CCS (∼0.120), FY4A-Official (∼0.148), and FY4B-Official (∼0.157). In summary, PECAM-FY4A&B consistently outperforms other precipitation products based on the CC, RMSE, and CSI metrics, with minimum improvements of approximately 0.046 in CC, 0.30 mm/h in RMSE, and 0.033 in CSI. In addition, FY4A-Official and FY4B-Official surpass CCS in all measures except RMSE.

D. Evaluations of PECAM-FY4A&B, CCS, FY4A-Official, and FY4B-Official Based on Global Daily Gauges

In the daily evaluation process, gauge records from NOAA are used as reference data (true values, details in Section II-B4). The precipitation pixels containing the gauges' locations are identified, and the gauge data within those pixels are averaged to correspond to the satellite footprints for direct comparison. Therefore, the pixels with no corresponding gauges are made transparent to better show the spatial distribution of the gauges. Similar to the hourly evaluation, CC, RMSE, and CSI are applied to show the overall performances of PECAM-FY4A&B, comparing with CCS, FY4A-Official, and FY4B-Official. Due to the uneven limitations of the spatial resolution of sample points, this evaluation is conducted using daily series at 0.1° resolution. In addition, the threshold for identifying precipitation events for CSI is set at 2.4 mm/day.

Based on Fig. 10, it is evident that CC highlights distinct geographical characteristics across all IR precipitation products, and they all demonstrate strong performance in the following areas: eastern China, regions around the equator, and the northern regions of Australia, where CC can reach 0.4 or more. The median correlation coefficient (CC) of PECAM-FY4A&B, at approximately 0.262, shows an improvement of at least 0.041 over the values of PERSIANN-CCS (∼0.221), FY4A-Official (∼0.202), and FY4B-Official

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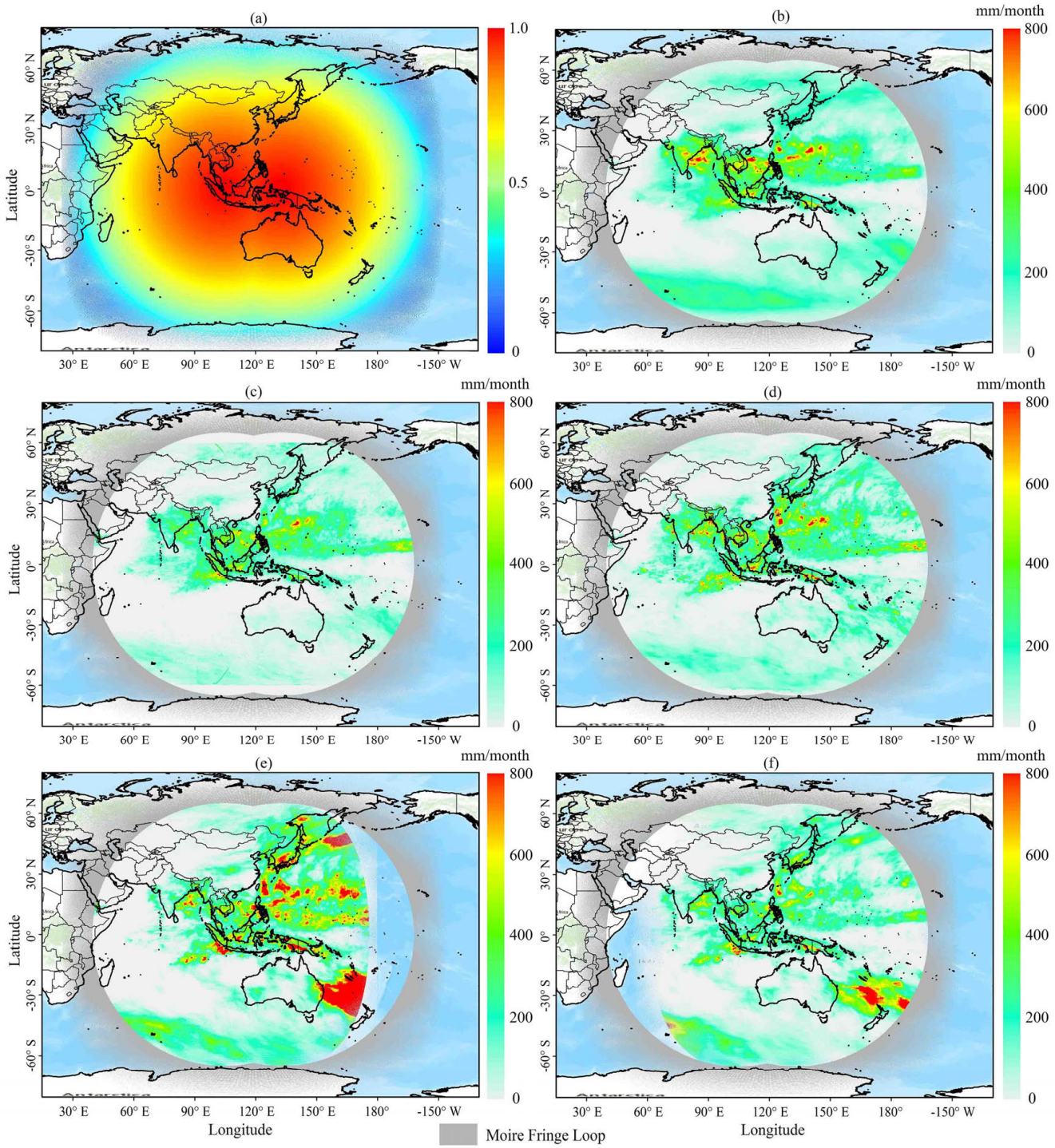


Fig. 8. Comparisons for spatial patterns of (a) max static VZA weights of FY-4A and FY-4B (representing the distance from the subpoints), (b) PECAM-FY4A&B, (c) PERSIANN-CCS, (d) IMERG, (e) FY4A-Official, and (f) FY4B-Official for September 2022.

(∼0.208), although this represents only marginal improvements. This limited enhancement can be attributed to the daily temporal resolution, which significantly narrows the gap in CC metrics relative to hourly scales. In terms of RMSE (improving at least 0.80 mm/day), both PECAM-FY4A&B (∼6.32 mm/day) and PERSIANN-CCS (∼7.12 mm/day) exhibit advantages when compared with FY-4 series data (∼11.95 and 11.57 mm/day for FY-4A and FY-4B, respectively), particularly in regions such as the Korean Peninsula, Japanese islands, and the eastern part of Australia. In these areas, the RMSE values of FY-4 series precipitation data mostly exceed 20 mm/day. When considering the capability of distinguishing precipitation events (in terms of CSI), PECAM-FY4A&B (∼0.272) also slightly outperforms PERSIANN-CCS (∼0.236), FY4A-Official (∼0.240), and FY4B-Official (∼0.250), improving at least 0.022. Notably, this difference in CSI is more pronounced over Indian and Southeast Asian regions. While PERSIANN-CCS, FY4A-Official, and

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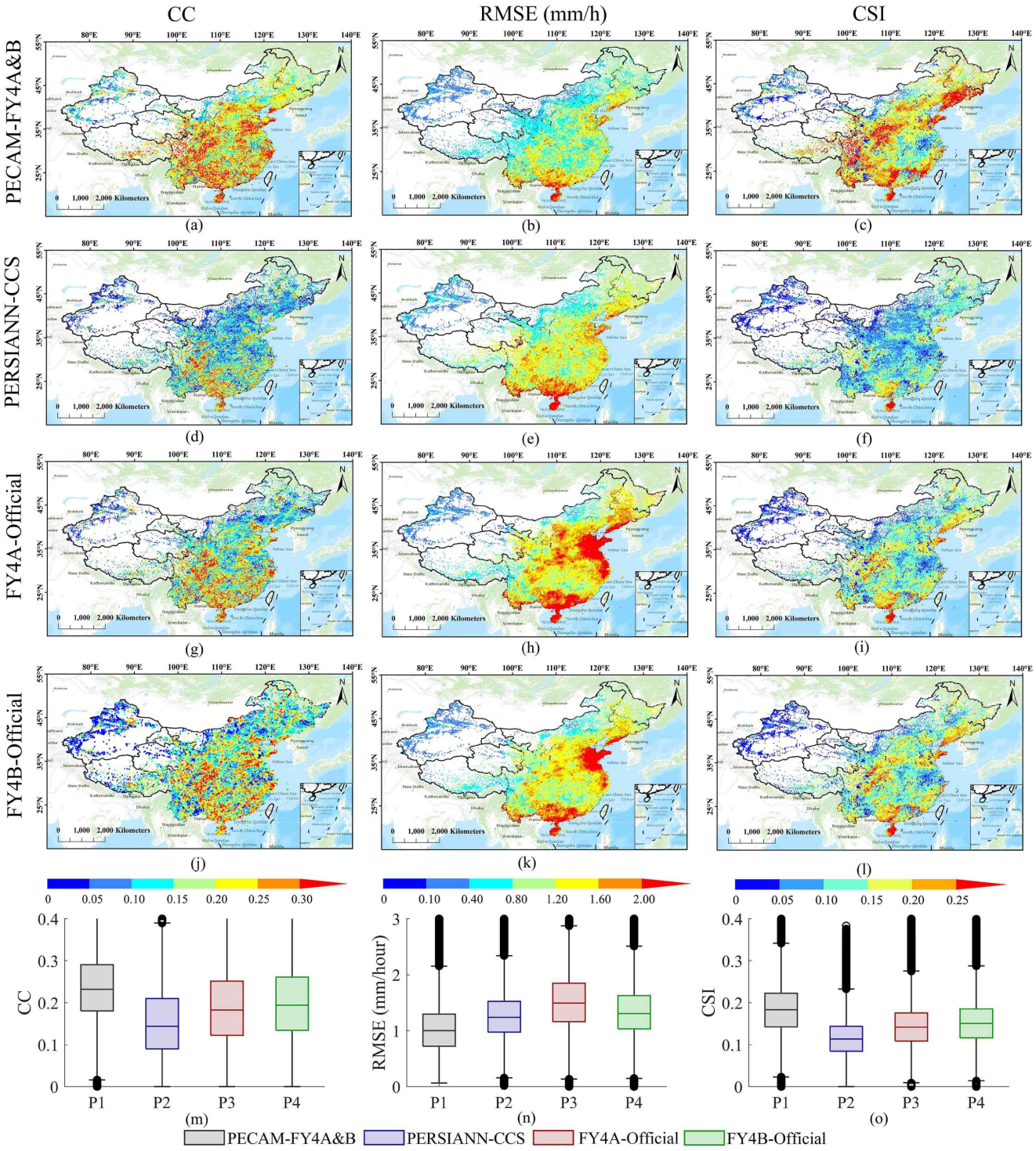


Fig. 9. Evaluation results using Chinese hourly gauge records as reference data for the period from June to September 2022. The figure presents the spatial distribution of metrics for (a)–(c) PECAM-FY4A&B, (d-f) PERSIANN-CCS, (g)–(i) FY4A-Official, and (j)–(l) FY4B-Official. Panels (m)–(o) summarize the statistical distribution of these metrics for all datasets. Note that the first, second, and third rows represent CC, RMSE, and CSI, respectively.

FY4B-Official demonstrate their respective strengths in various aspects, the substantial enhancements in RMSE highlight that PECAM-FY4A&B is better suitable for long-term precipitation analysis at a daily scale from 60°E to 180°E.

E. Enhanced Feature Performances During a Typical Precipitation Event

To better assess the effectiveness of feature enhancements, we focus on a typical precipitation event occurring on September 28, 2022, at 14:00 UTC, within the regions spanning from 19°N to 29°N and from 127°E to 137°E. Fig. 11(a) and (b) shows the precipitation data named PECAM-FY4A and PECAM-FY4B, which are generated using feature enhancement strategies based on observations from FY-4A and FY-4B, respectively. Using IMERG as our reference data [see Fig. 11(e)], we can identify two prominent precipitation centers for this event. The first center is situated around 23.7°N and 131.4°E, and the second

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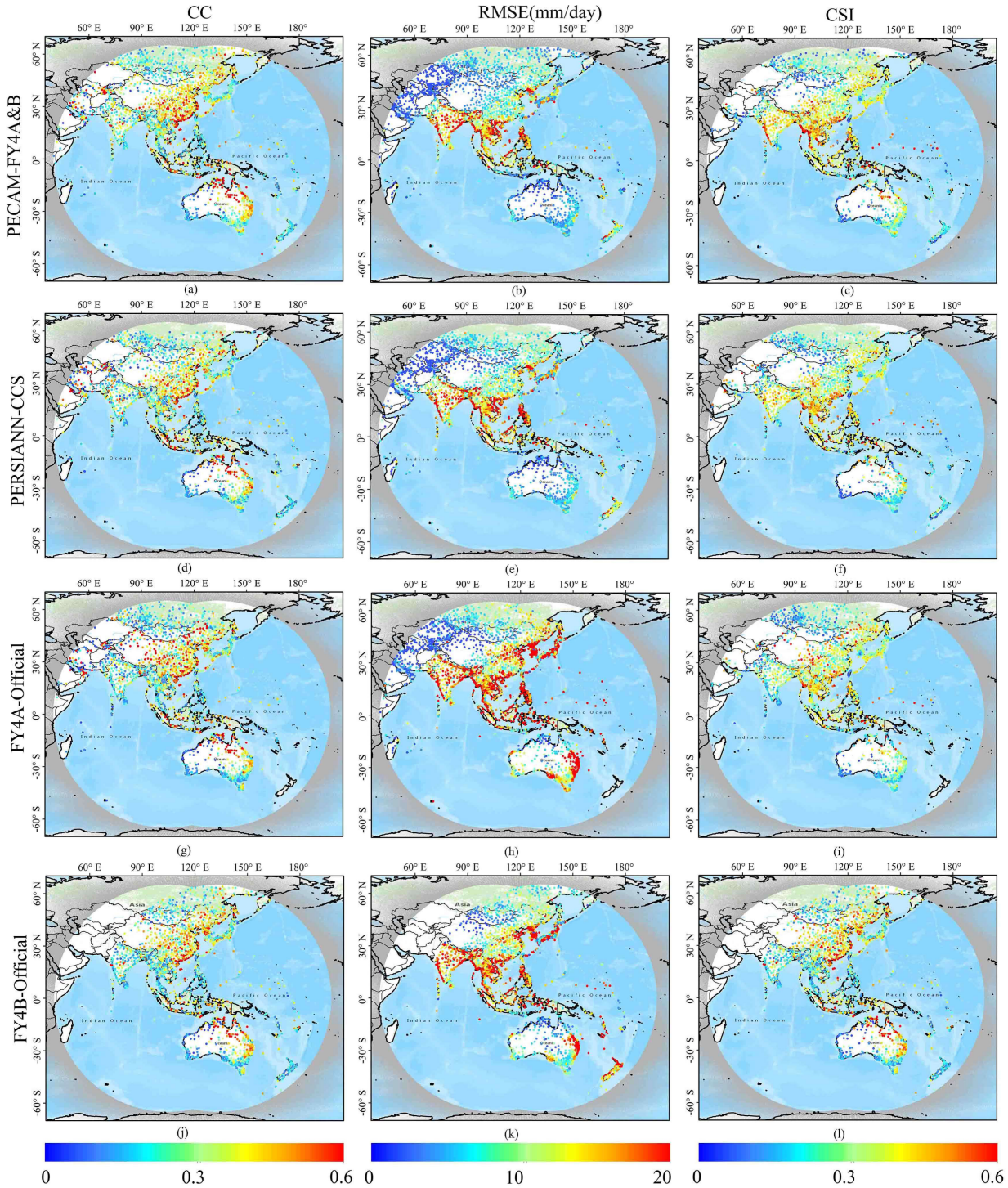


Fig. 10. Evaluation results using global daily gauge records as reference data from June to September of 2022. The figure presents the spatial distribution of metrics for (a)–(c) PECAM-FY4A&B, (d)–(f) PERSIANN-CCS, (g)–(i) FY4A-Official, and (j)–(l) FY4B-Official. Note that the first, second, and third rows represent CC, RMSE, and CSI, respectively.

center is approximately at 24.6°N and 132.3°E. However, when examining PERSIANN-CCS [see Fig. 11(c)], it displays an additional precipitation center at roughly 24.5°N and 133.2°E, which is evidently a misjudged precipitation center. From Fig. 11(d), FY4A-Official significantly overestimates the precipitation areas in these two centers. Meanwhile, FY4A-Official employs a strategy of capping all precipitation values exceeding 20 mm/h to avoid extreme precipitation values. Consequently, when we analyze the precipitation values along the particular line of interest [see Fig. 11(f)], FY4A-Official fails to capture the extreme precipitation center with values exceeding 20 mm/h, and these values are replaced with a flat and featureless representation. In addition, all IR precipitation products in this study appear to notably overestimate precipitation around 23.7°N and 130.8°E when detecting precipitation centers toward the southeast. This overestimation may be

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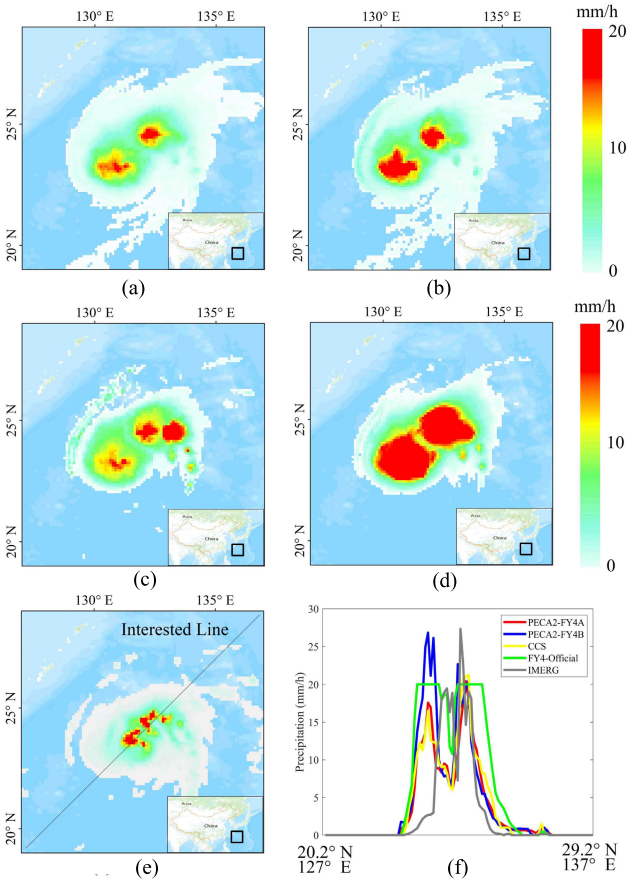


Fig. 11. Precipitation distributions in a typical extreme precipitation event (detailed information), including (a) PECAM-FY4A, (b) PECAM-FY4B, (c) PERSIANN-CCS, (d) FY4A-Official, (e) IMERG, and (f) precipitation curves along an interested line (from 19.2°N and 127°E to 29.2°N and 137°E).

attributed to the challenges IR observations face in delineating precipitation bands beneath the central dense overcast of a tropical system. This deviation is likely due to the constraints of IR signals in retrieving information under thick cloud cover, while IMERG performs better in circular storms because it considers microwave rain rates.

V. Discussion

A. Necessity of Feature Enhancements for IR Signals

In the original PECA structure, IR band features were quantified directly to establish a relationship with precipitation values using a statistical method, which drew criticism for neglecting the impact of solar radiation. Therefore, PECAM has taken into account more factors that could introduce systematic variations in band features, including latitude, VZA, elevation, SZA, EOA, and shadow effects. However, much of the radiative transfer processes relevant to precipitation cannot be fully characterized using IR data alone, as precipitating clouds are optically thick and absorb most radiation below the cloud top [70]. Therefore, PECAM can only provide approximations of the influence of geographical features and the contributions of certain band features. Exploring the IR radiative transfer processes of precipitation requires additional work and will be a focus for future research.

To address this challenge, chromatographical theory has been applied to simulate the dynamic changes in IR signals with varying precipitation levels. This strategy involves customizing a function with adjustable parameters in a chromatographical neuron to better fit the patterns between precipitation and band features [4]. Furthermore, a feature enhancement process has been designed to highlight the aspects related to precipitation while minimizing noise. During this process, systematic variations in IR band features, such as diurnal patterns influenced by the SZA, are mitigated. This results in more accurate mapping curves within the chromatographical neuron, enabling a more precise capture of precipitation patterns.

B. Analyzing IR Band Feature Performances Through Radiation Transformation

The aforementioned geographical factors exert various influences on different band features, resulting in both systematic and random errors in precipitation estimation [64], [68], [71]. Among these factors, latitude, VZA, EOA, and elevation would contribute to systematic errors in precipitation estimation, while only the shadow effect of clouds would contribute to random errors. We observed that even in the absence of precipitation, regions with higher latitudes, greater elevations, and larger VZA often exhibit lower TBBs, potentially leading to precipitation overestimation when they follow the same precipitation—TBB curves as other regions. The brightness temperature effects in clear skies significantly differ from those in cloudy conditions, highlighting the need to investigate the underlying mechanisms of the radiative transfer process in future studies [61]. Since optically thick precipitating clouds absorb virtually all surface radiation, the variations in temperature at cloud level with latitude and changes in cloud-top heights, rather than changes in surface emissions, significantly impact the TBBs in IR observations [72], [73], [74]. These factors are the primary contributors to the systematic errors in IR-based precipitation estimation associated with geographical influences.

The SZA has a complex impact on band features, partly due to its dependence on the wavelength of bands. As a result, the peaks of solar emissions are predominantly in the visible wavelength range, while Earth's surface emissions are concentrated in the IR wavelength range [75]. This distinction means that solar radiation has a more pronounced influence on bands with wavelengths closer to visible light, particularly on $T_{3.75H}$ and $T_{3.75L}$. Meanwhile, SZA primarily affects the scattering and reflection of solar radiation by clouds and the Earth's surface, leading to significant diurnal variations in $T_{3.75H-13.5A}$ [76]. In addition, solar radiation significantly increases $T_{3.75H}$ and $T_{3.75L}$, causing $T_{3.75H-3.75L}$ to decrease during daylight hours.

Instead of diurnal fluctuations, the EOA reveals significant seasonal variations in $T_{7.1A}$ and $T_{13.5A}$, which are sensitive to water vapor and air temperature, respectively. Consequently, $T_{7.1A-13.5A}$, representing the disparity between water vapor bands and far-IR bands, exhibits distinct seasonal patterns closely associated with EOA. For instance, $T_{7.1A-13.5A}$ displays relatively higher values (>8 K) in the Southern Hemisphere during the summer, while similarly high values (>8 K) are observed in the NH during the winter. This phenomenon can be attributed to the combined effects observed during the cold

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seasons in specific regions, where higher $T_{7.1A}$ and lower $T_{13.5A}$ result in an increased $T_{7.1A-13.5A}$. It is important to note that this should be attributed to the synergistic impact of both the warming effect of reduced water vapor and the cooling effect of a colder tropospheric temperature profile. Therefore, the primary radiative processes influencing $T_{7.1A-13.5A}$ involve the emissions from the Earth's surface, interactions with the surrounding air, and the absorption of water vapor in the atmosphere [77].

The shadow effect is a unique source of random error that specifically manifests in $T_{3.75H-3.75L}$. This effect is characterized by the presence of extremely low values (below -3 K) occurring alongside high values (above 5 K), which often indicate the presence of precipitation. Upon visual examination of these shadows (regions where $T_{3.75H-3.75L} < -3$ K), it becomes evident that they tend to appear on the land surface directly beneath clouds (regions where $T_{3.75H-3.75L} > 5$ K). These shadow patterns are closely tied to the SZA. These shadows disrupt the continuity of mapping curves in chromatographical neurons and introduce noisy estimates into precipitation data. To mitigate this issue and preserve cloud-related information, we have chosen to cap negative values of $T_{3.75H-3.75L}$ at zero. This action effectively removes the shadows while retaining cloud-related features. This decision is grounded in the understanding that $T_{3.75H}$ and $T_{3.75L}$ primarily reflect the scattering and reflection processes of clouds and land surfaces. Furthermore, $T_{3.75L}$ is particularly sensitive to heavy precipitation or thick cloud cover [4]. Consequently, over regions with cloud cover, $T_{3.75L}$ tends to be lower, resulting in higher values for $T_{3.75H-3.75L}$, which serves as an indicator of extreme precipitation. In addition, for the land areas directly under the clouds, sections not covered by precipitation or clouds exhibit higher $T_{3.75L}$ values, leading to lower $T_{3.75H-3.75L}$ values.

VI. Conclusion

Utilizing multiple spectral IR observations from geostationary satellites, such as FY-4A and FY-4B equipped with AGRI, offers highly effective high-resolution precipitation estimation in eastern Asia. Nevertheless, geographic factors, including VZA and SZA, introduce systematic errors into the estimation and merging processes. In response to this challenge, the PECAM-FY4A&B was developed. The primary findings can be summarized as follows.

1) The average TBB demonstrates a negative correlation with latitude, VZA, and elevation, especially near boundaries, with reductions of up to 20 K, causing precipitation overestimation.
2) EOA strongly influences the water vapor band minus longwave IR band (as $\Delta T_{7.1A-13.5A}$ and $\Delta T_{6.95B-13.3B}$), leading to pronounced seasonal variations.
3) Because the near-IR band is sensitive to the SZA, it ultimately results in conspicuous diurnal fluctuations in $\Delta T_{3.75H-13.5A}$ and $\Delta T_{3.75H-13.3B}$.
4) Shadow effects are primarily observed in the $\Delta T_{3.75H-3.75L}$ band, manifesting as shadow effects (<-3 K) beneath precipitation clouds (>5 K).
5) At hourly scale, PECAM-FY4A&B consistently outperforms other precipitation products across CC, RMSE, and CSI metrics, with minimum improvements of approximately 0.046 in CC, 0.30 mm/h in RMSE, and 0.033 in CSI, respectively.
6) At a daily scale with global gauge datasets as references, PECAM outperforms PERSIANN-CCS, FY4A-Official, and FY4B-Official with at least a 0.041 increase in CC, a 0.80-mm/day reduction in RMSE, and a 0.022 increase in CSI, respectively.

These findings underscore the significant potential of the signal processing and merging strategy employed in PECAM, not only as a reference for improving precipitation estimation from the FY-4 series but also for enhancing data integration from the GOES, Meteosat, and Himawari series. The advancement in understanding and mitigating the impact of geographic factors on precipitation estimation is a noteworthy achievement, promising more accurate and reliable weather data for a variety of applications.

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